❄️ Snowflake Learning Series – SnowSQL (End-to-End Explanation) ❄️

What is SnowSQL?

SnowSQL is a **Python-based Command Line Interface (CLI)** used to connect to **Snowflake** from operating systems like:

- Windows
- Linux
- macOS

It allows users to interact with Snowflake directly from the **terminal or command prompt**.

Using SnowSQL we can:

- Execute SQL queries
- Create database objects (tables, schemas, warehouses)
- Perform DDL & DML operations
- Load and unload data
- Run scripts for automation
- Manage Snowflake environments

In simple terms:

👉 **SnowSQL = Terminal tool to communicate with Snowflake.**

Ways to Access Snowflake

Snowflake can be accessed through multiple interfaces depending on the use case.

1️⃣ Snowflake Web Interface

Browser-based UI used for:

- Writing SQL queries
- Managing databases
- Monitoring warehouses
- Viewing query history

Snowflake recommends using **Google Chrome**.

2️⃣ SnowSQL (Command Line Client)

A terminal-based tool used mainly by:

- Data Engineers
- Developers
- Automation pipelines

Example usage:

```
snowsql -a account_name -u username
```

3 JDBC / ODBC Drivers

Used by applications to connect to Snowflake.

Examples:

- Java applications → JDBC
- BI tools → ODBC

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4 Third-Party Tools

Many tools integrate directly with Snowflake.

Examples:

- Tableau
- Matillion
- Power BI
- Informatica
- Fivetran

Installing SnowSQL

SnowSQL can be downloaded from the **Snowflake website**.

The installer includes all required dependencies.

Supported Operating Systems

Windows
- Windows 7 or later
- Windows Server 2008 R2 or later

macOS
- Version 10.12 or later

Linux
- CentOS 6+
- Ubuntu 14+

Installation Steps

1️⃣ Login to Snowflake Web UI

2️⃣ Navigate to

Help → Downloads

3️⃣ Download **SnowSQL CLI** for your OS.

4️⃣ Run the installer.

5️⃣ Follow the installation wizard.

6️⃣ After installation you will see

Installation is Complete

Now SnowSQL is ready to use.

Configuring SnowSQL

After installation, SnowSQL must be configured using the **config file**.

Configuration file name:

config

Location:

Linux / macOS

~/.snowsql/

Windows

C:\Users\<username>\.snowsql\

Important rule:

👉 The config file must be saved in **UTF-8 encoding**.

SnowSQL Configuration Sections

The configuration file has **three main sections**:

1️⃣ **Connection Settings**

Stores account login details.

2️⃣ **Configuration Options**

Stores CLI settings.

3 Variables

Stores reusable variables

Connection Settings

To connect SnowSQL to your Snowflake account, you need:

- Account Name
- Username
- Password
- Region

Snowflake URL Format

AWS (US-West)

https://<account_name>.snowflakecomputing.com

AWS (Other Regions)

https://<account_name>.<region>.snowflakecomputing.com

Azure

https://<account_name>.<region>.azure.snowflakecomputing.com

Example:

https://ru06473.ap-southeast-1.snowflakecomputing.com

Testing SnowSQL Connection

Use the following command:

snowsql -a account_name -u username

Example

snowsql -a ru06473.ap-southeast-1 -u SKILLSCALER

If successful, SnowSQL will open an interactive session.

Saving Credentials in Config File

To avoid typing credentials every time, edit the config file.

Example configuration:

accountname = wr82310
username = SKILLSCALER1
password = xxxxxxxx
region = ap-south-1

Then simply run:

snowsql

⚠ Important:

Passwords are stored in **plain text**, so the config file must be **secured properly**.

SnowSQL Variables

Variables allow you to store frequently used values like:

- database name
- schema name
- table name
- date values

This improves productivity.

Example variable definition:

[variables]

database_name = SNOWFLAKE_SAMPLE_DATA
schema_name = TPCH_SF001
table_name = NATION

Using Variables in SnowSQL

Start SnowSQL:

snowsql

Enable substitution:

!set variable_substitution=true

Use variables:

USE "&database_name";
USE SCHEMA "&schema_name";

SELECT COUNT(*) FROM "&table_name";

Command Line Variables

Variables can also be defined when starting SnowSQL.

Example:

snowsql -D tablename=NATION -s TPCH_SF001 -d SNOWFLAKE_SAMPLE_DATA

Then run:

```
!set variable_substitution=true
select count(*) from "&tablename";
```

Variables in Active Session

Variables can also be defined inside an active session.

Example:

```
!define tablename=NATION
!set variable_substitution=true
SELECT COUNT(*) FROM "&tablename";
```

SnowSQL Commands

SnowSQL commands start with **!**

Example:

!help

This displays all available commands.

Common SnowSQL Commands

Abort Query

!abort <query_id>

Create Connection

!connect <connection_name>

Define Variable

!define variable=value

Open SQL Editor

!edit

Show Options

!options

Exit Session

!exit

Quit SnowSQL

!quit

Difference:

!exit → closes one connection
!quit → closes all connections

Execute SQL File

!source filename.sql

Save Query Output to File

!spool output.txt

Stop writing output

!spool off

Multiple Connections in SnowSQL

SnowSQL supports connecting to **multiple environments**.

Example environments:

- Development
- Testing
- Production

These can be configured in the config file.

Example:

[connections.development]
password=xxxx
warehousename=DEVELOPMENT

[connections.production]
password=xxxx
warehousename=PRODUCTION

Creating Warehouses for Environments

Example:

CREATE WAREHOUSE DEVELOPMENT
WAREHOUSE_SIZE='XSMALL'
AUTO_SUSPEND=600
AUTO_RESUME=TRUE;

CREATE WAREHOUSE PRODUCTION
WAREHOUSE_SIZE='XSMALL'
AUTO_SUSPEND=600
AUTO_RESUME=TRUE;

Switching Connections

!connect development
!connect production

Connection stack example:

Development → Production → Exit → Back to Development

Data Loading Using SnowSQL

SnowSQL can be used to **automate data pipelines** using the COPY command.

Example file:

zips2000.csv

Step 1 – Set Environment

USE WAREHOUSE COMPUTE_WH;
USE DATABASE DEMO_DB;
USE SCHEMA PUBLIC;

Step 2 – Create Table

```
CREATE TABLE ZIPCODES2000_SNOWSQL
(
ZIPCODE STRING,
LON DOUBLE,
LAT DOUBLE
);
```

Step 3 – Upload File to Stage

```
PUT file:///Users/path/zips2000.csv
@DEMO_DB.PUBLIC.%zipcodes2000_snowsql;
```

Step 4 – Load Data

```
COPY INTO zipcodes2000_snowsql
FROM @%zipcodes2000_snowsql
FILE_FORMAT = (
TYPE = CSV
FIELD_OPTIONALLY_ENCLOSED_BY='"'
SKIP_HEADER = 1
);
```

Step 5 – Verify Data

```
SELECT * FROM zipcodes2000_snowsql;
```

SnowSQL Connection Command

Basic syntax:

```
snowsql -a accountname -u username
-d database
-s schema
```

Example:

```
snowsql -a xta99637.us-east-1
-u vithaljs
-d demo_db
-s public
```

Common SnowSQL Command Options

Option	Description
-a	Account name
-u	Username
-d	Database
-s	Schema
-r	Role
-w	Warehouse
-q	Execute query
-f	Run SQL file
-D	Define variable
-c	Use connection
-v	Show version
--help	Show help

Real-Time Use Case

In real projects SnowSQL is used for:

- Automating data pipelines
- Scheduling ETL jobs
- Running batch scripts
- Loading large files into Snowflake
- Running CI/CD data workflows

Example:

snowsql -f load_data.sql

This runs the entire pipeline automatically.
